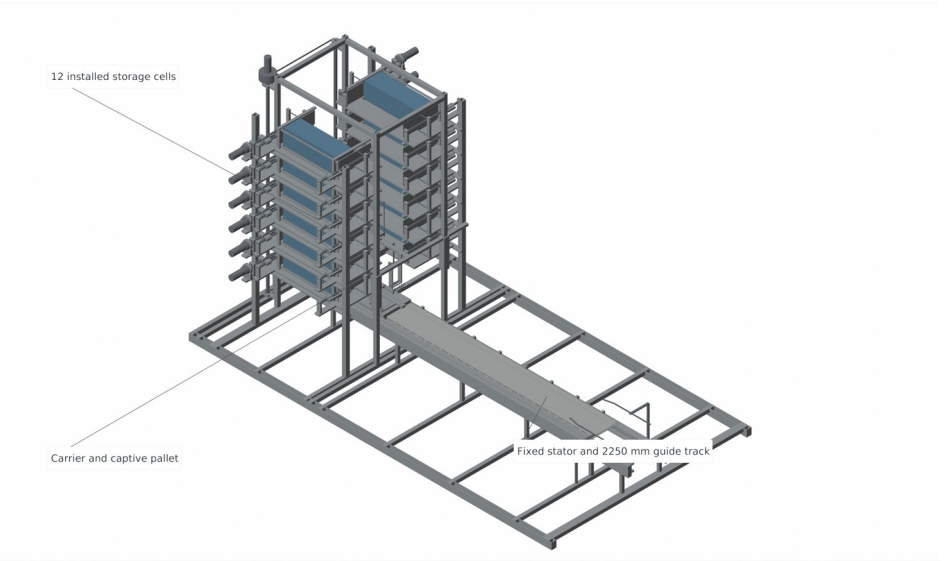
DEPARTMENT OF MECHANICAL ENGINEERING

VOLLEY

Design and Computational Evaluation of a Commanded Electromagnetic CubeSat Deployment Mechanism

GEN5 ACR1 | COMPUTATIONAL DESIGN STUDY



Adityavardhan Mishra
PRN: 23070125054

Pratham Chawla
PRN: 23070125029

Project guide: Vikas Gulia

Symbiosis Institute of Technology, Pune

Presentation outline

01 Motivation and prior work Introduction, literature, research gap and objectives

02 Method and progress Controlled configuration and dependent numerical models

03 Mechanism and results Payload custody, finite motor, source, structure and mission

04 Conclusions and next steps Study outcome, Gen6 prototype and references

Commanded release adds a local delivery variable

DEPLOYMENT EVENT - SCHEMATIC



Concept only; arrows are not to scale.

- Host manoeuvres establish the carrier state.
- A release command changes the payload initial condition relative to the host.
- Retention, guidance, arrest and host recoil complete the system boundary.

Research question

Can a twelve-payload system complete a coupled, reusable deployment cycle?

Sources: JAXA deployment ICD; Cal Poly CDS Rev. 14.1; Foster et al. (2018); Vallado (2013)

Prior work establishes the mechanisms and comparison context

Published work	Evidence / interface	Implication for VOLLEY
Zhao et al., 2022 Stacked EM deployment	Gravity-compensated tests; stacked CubeSat release	Distinguish published measurements from the present computational results.
Zhao et al., 2025 High-volume storage	Receiving guides and magnetic transfer; payload grooves and iron patches	Define the passive payload interface and receiving custody explicitly.
Feng et al., 2025 EM reachable domain	Higher-speed launcher and mission-domain analysis	Couple the selected operating regime to its energy and mission assumptions.
Halbach; finite-motor studies	Permanent-magnet field shaping and longitudinal end effects	Use finite magnet and conductor geometry at launch and arrest.

Study focus: integration and same-revision evaluation of a retained-carrier architecture.

Sources: Zhao et al. (2022, 2025); Feng et al. (2025); Halbach (1980); Yamaguchi & Morishita (2023)

Research gap: closing the complete retained-carrier cycle

Who holds the payload?

A complete custody path through cell opening, receiving, docking and commanded departure.

Can the source deliver it?

Finite force, actual moving mass, current dynamics and post-release arrest evaluated together.

What does the host retain?

The reusable carrier, installed hardware, power allowance and recoil carried into mission comparison.

One controlled design links mechanical interfaces, electromechanics and mission resources.

Objectives and controlled reference requirements

12 × 4 kg
Identical 3U envelope surrogates

1–3 m/s
Analysed commanded-speed cases

2250 mm
Guide-rail allocation

210 mm
Post-release arrest allocation

- Define continuous payload custody and a reusable carrier cycle.
- Evaluate finite motor force with the represented winding and magnet geometry.
- Couple source, release and arrest; assess structural and mission consequences.

Sources: frozen CAD manifest; PARAMETERS.json; coupled operating-study definition

Methodology follows the physical dependency chain

1 — Controlled CAD
Solids and custody states

2 — Mass and inertia
Solid ledger and allowances

3 — Finite motor
Field and current paths

4 — Coupled shot
Source and release/arrest

5 — Mechanical budgets
Chassis and interfaces

6 — Mission boundary
Release map and host mass

Computational cross-checks

Geometry

Native / STEP readback

Electromagnetics

Source-face and current-FEM checks

Dynamics

5 µs timestep audit (shared solver)

Sources: engineering manifests; finite field/current models; STEP readback and reviewed 5 µs timestep-audit records

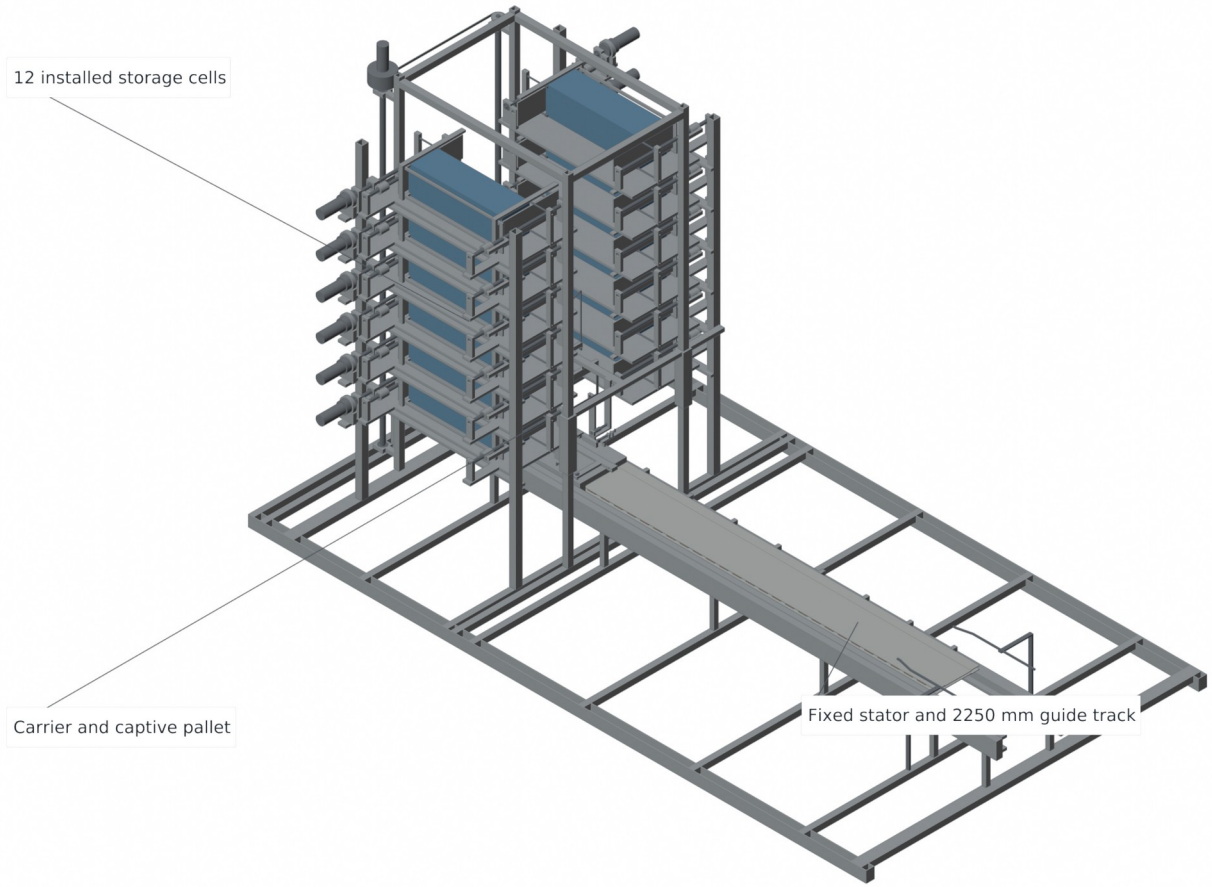
Timeline and work progress

01	02	03	04	05
Problem and literature	**Gen5 ACR1 definition**	**Coupled computational study**	**Report and presentation**	**Gen6 instrumented prototype**
Complete	Complete	Complete	Current submission	Future work

Completed outcome

A controlled computational design reference with defined operating cases and a hardware validation programme.

Reference architecture: fixed banks and one captive pallet



2 × 6 cells
Fixed, individually retained storage

13.128 kg
Retained moving carrier + pallet

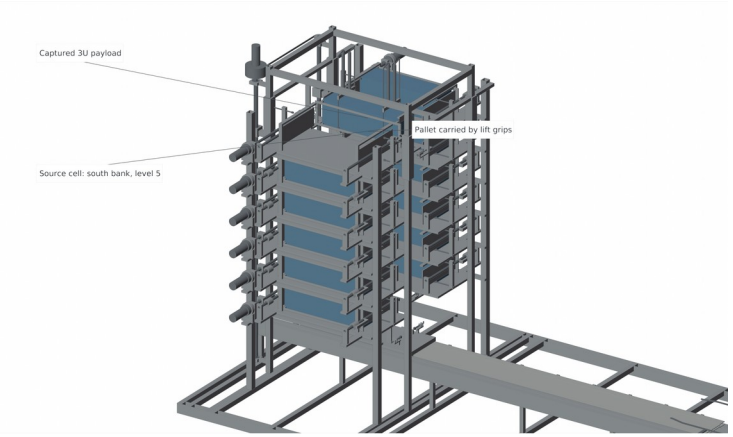
149.557 kg
Represented dry CAD mass

+7 kg power/control allowance

Magnet arrays stay on the carrier.

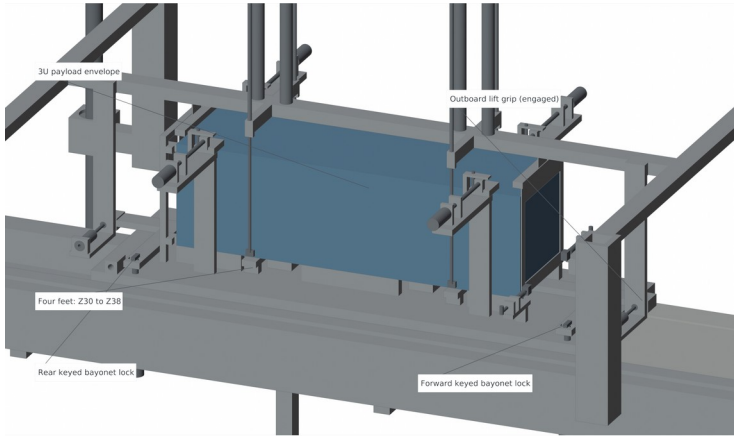
Native CAD render: final Gen5 ACR1 home state; represented masses include declared reference components

Capture before release: three explicit custody states



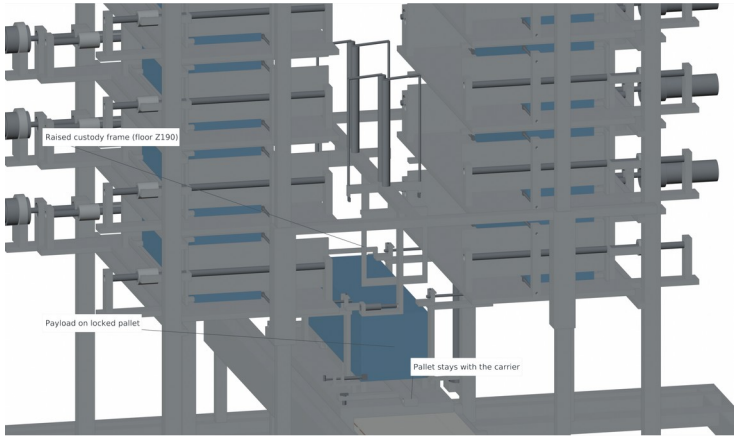
1 Capture at the source

Lift grips support the pallet; end frames and Y keepers contain the payload.



2 Dock and lock

Carrier caps and both bayonets engage before the outboard lift grips open.



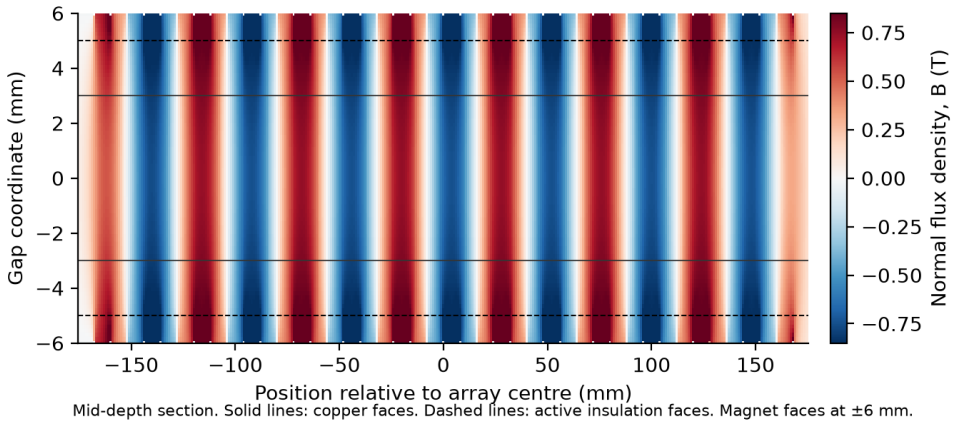
3 Raise clear

The pallet stays on the carrier through release, arrest and return.

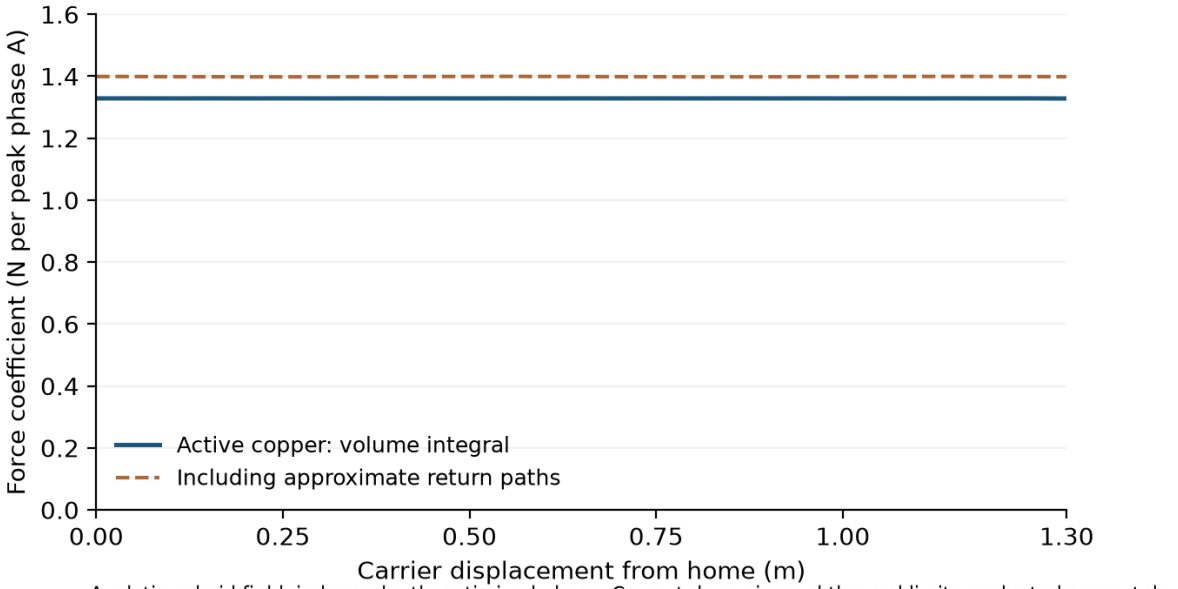
Passive cam: 12 mm forekeeper opening over 50 mm travel, complete 5 mm before release.

Native CAD states; handoff detail hides banks, host ladder and fixed lift frame as disclosed in its provenance

Finite motor geometry enters the force model



Mid-depth section. Solid lines: copper faces. Dashed lines: active insulation faces. Magnet faces at ±6 mm.



Analytic cuboid field; independently optimized phase. Current dynamics and thermal limits evaluated separately.

42 + 42 magnets
8 mm blocks on opposed retained arrays

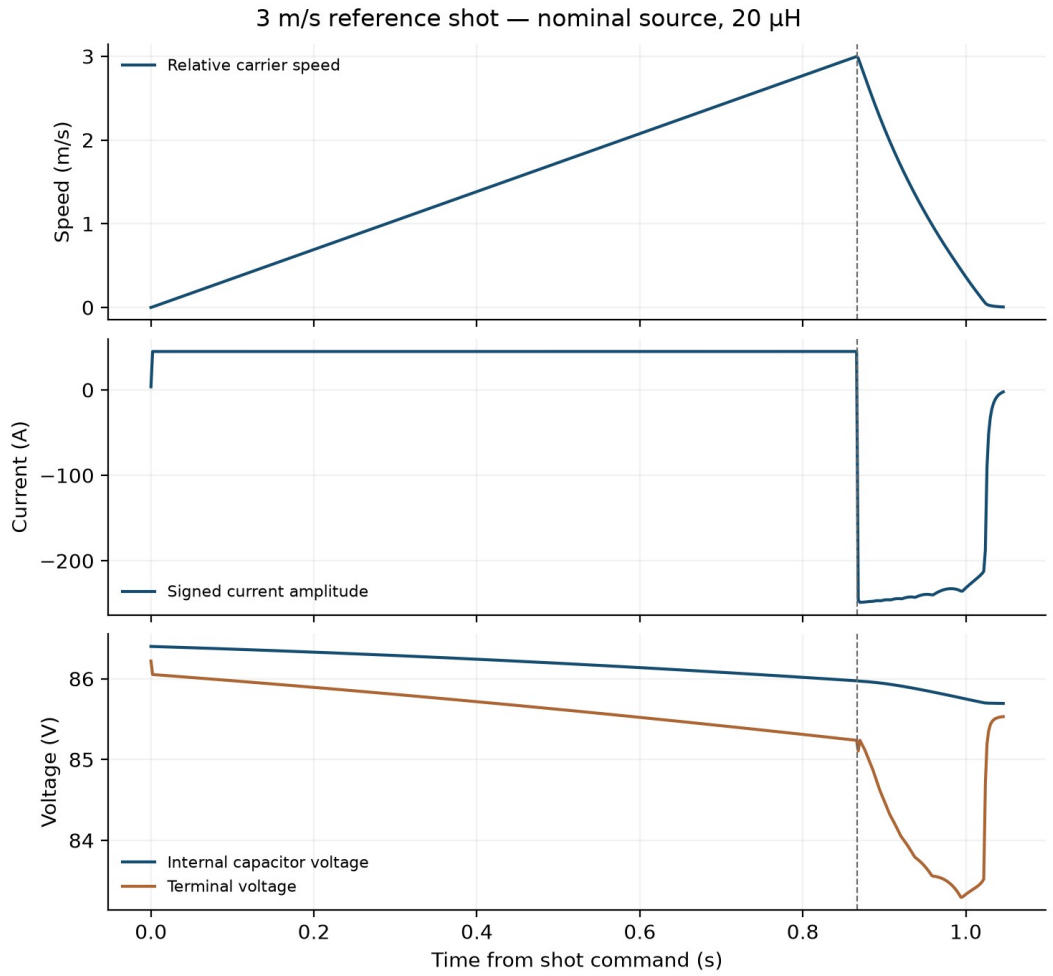
216 active bars
42 mm² bars; 21 mm² return necks

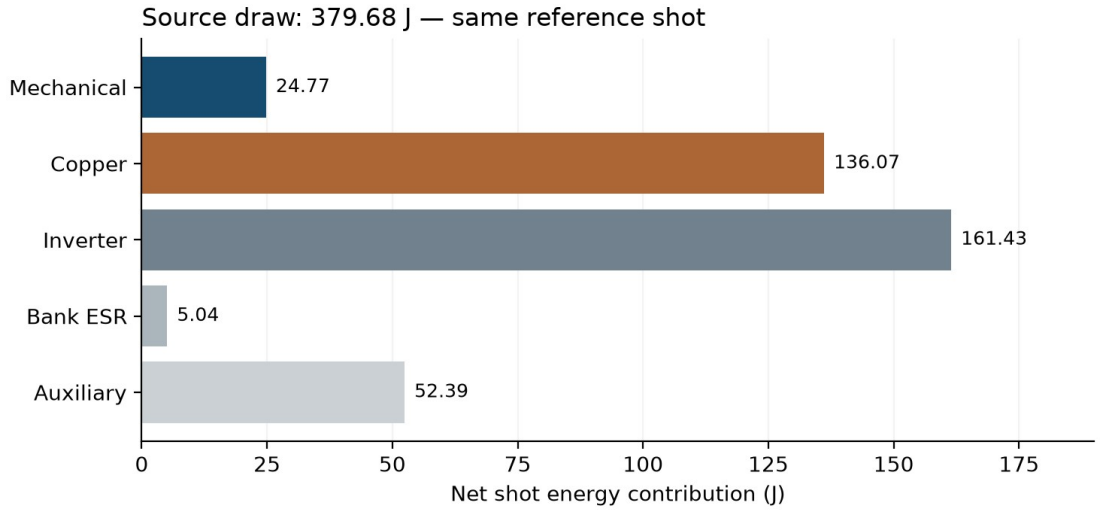
0.172%
Interior current-FEM force difference

At x = 0.65 m: active 1.3287 N/A; represented returns included 1.3991 N/A.

Sources: FINITE_MOTOR.json; CURRENT_FIELD_FORCE.json | coefficients use peak phase amperes

Coupled shot: release, source and active arrest





3.000 m/s

Model release speed

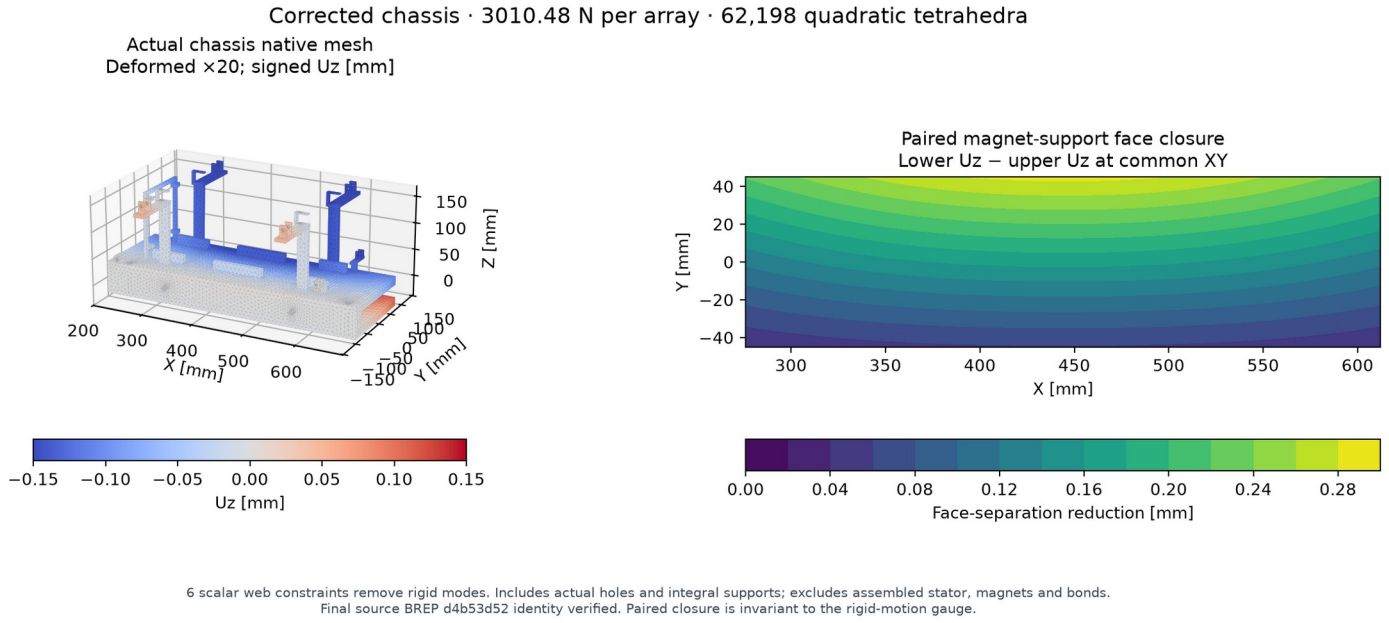
379.7 J

Nominal source draw

6/6 cases pass the 5 µs timestep audit

Worst reviewed single-shot adiabatic estimates:
Bulk ≤0.0667 K; local DC-FE mesh observation 2.495 K.

Mechanical evaluation: actual chassis and declared contact budgets



Linear-elastic FEA; 3.010 kN per array; displacement shown ×20.

0.286 mm
Maximum paired face closure

0.372 mm
Declared 1.3× linear load sensitivity

Allocated per-face gap margin

0.264 mm

Four-rail contact surrogate

Elastic-energy rate bounds

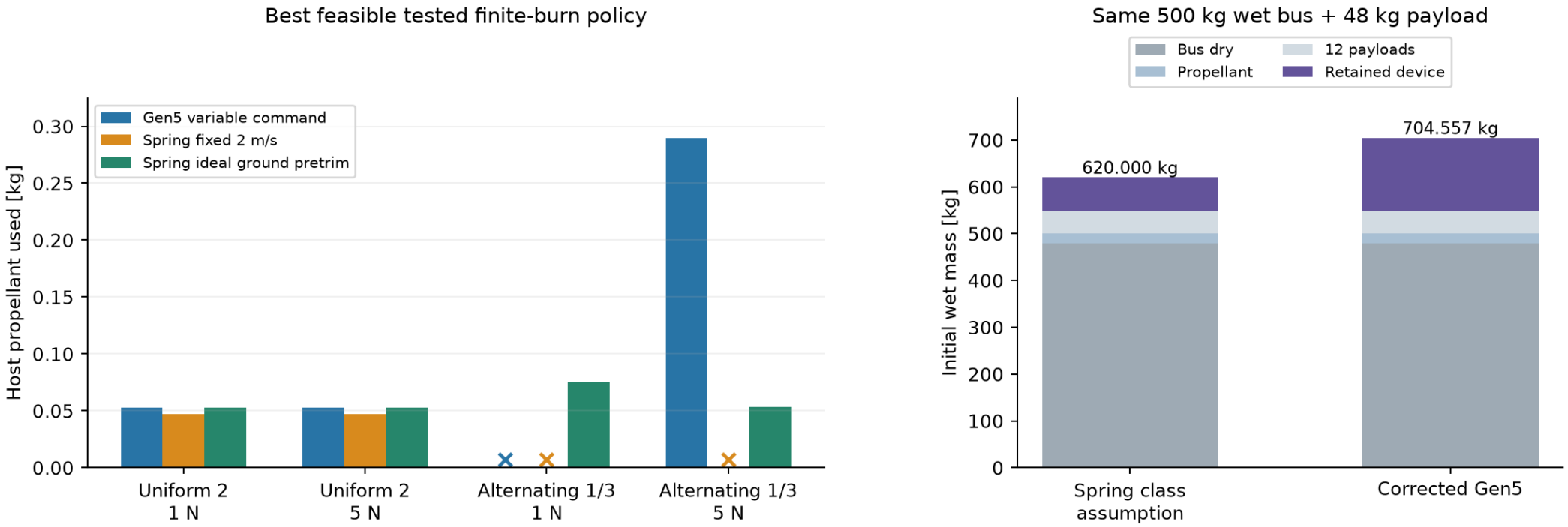
1.044°/s pitch/yaw

2.619°/s all-axis

Declared stiffness and inertia.
Gen6 physical correlation.

Sources: actual-chassis CalculiX solve; DESIGN_BUDGETS.json | simulations and analytical bounds

Mission trade: retain the complete mass and control boundary



× means no tested candidate met that force ceiling; it is not a global infeasibility result.
Gen5 device = 149.5567 kg CAD + 7 kg allowance; spring device = declared 12 × 6 kg class assumption.

Ideal pretrim uses less fuel and retained mass in this comparison.

Control families: Gen5 has 400 s coast-only releases; spring policies allow broader burns.

Conclusions of the computational design study

A complete reference cycle is defined

Fixed cells, guided feed, positive pallet capture, commanded release, active arrest and return.

The declared low-speed cases are computationally supported

1–3 m/s cases analysed; the principal 3 m/s matrix passes a reviewed shared-solver timestep audit.

The retained system burden is visible

149.557 kg represented dry CAD plus 7 kg allowance; no resource advantage over ideal pretrim in the stated mission comparison.

Gen5 ACR1 completes the computational reference for physical validation.

Gen6: an instrumented mixed 1U / 3U physical prototype

1 Characterise

Force / back emf

Resistance and inductance

Source ESR and switching

2 Correlate custody

Pusher and guide forces

Pallet and lock repeatability

Payload contact and angular rate

3 Demonstrate the cycle

Release and active arrest

Repeated thermal duty

Mixed-envelope handling

Further scaling requires fresh stroke, acceleration, energy, arrest and heat-rejection budgets.

Selected references

01 Cal Poly, CubeSat Design Specification, Rev. 14.1, 2022; JAXA, JEM small-satellite deployment ICD.

02 Zhao et al., “Design and analysis of a new deployer for the in-orbit release of multiple stacked CubeSats,” Remote Sensing 14, 4205, 2022. doi:10.3390/rs14174205.

03 Zhao et al., “Simulation analysis and experimental verification of the transport characteristics of a high-volume CubeSat storage device,” Aerospace 12, 466, 2025. doi:10.3390/aerospace12060466.

04 Feng, Yang and Wu, “Design and reachable domain analysis of on-orbit electromagnetic launcher for CubeSats,” IJAE, 3000765, 2025.

05 Halbach, “Design of permanent multipole magnets with oriented rare earth cobalt material,” NIM 169, 1–10, 1980.

06 Yamaguchi and Morishita, “End effects of Halbach field magnet-type linear motor,” Electrical Engineering in Japan, 2023. doi:10.1002/eej.23450.

07 Foster et al., “Constellation phasing with differential drag on Planet Labs satellites,” Journal of Spacecraft and Rockets 55(2), 2018.

08 Vallado, Fundamentals of Astrodynamics and Applications, 4th ed., 2013.

09 Component references: Eaton TV cells; Infineon IRFP4668PbF; Texas Instruments UCC21520; FAULHABER 3242 BX4 and 32GPT.

10 Methods: FreeCAD; Magpylib (SoftwareX 11, 100466); Gmsh (IJNME 79, 1309–1331); CalculiX; Blender.

Questions